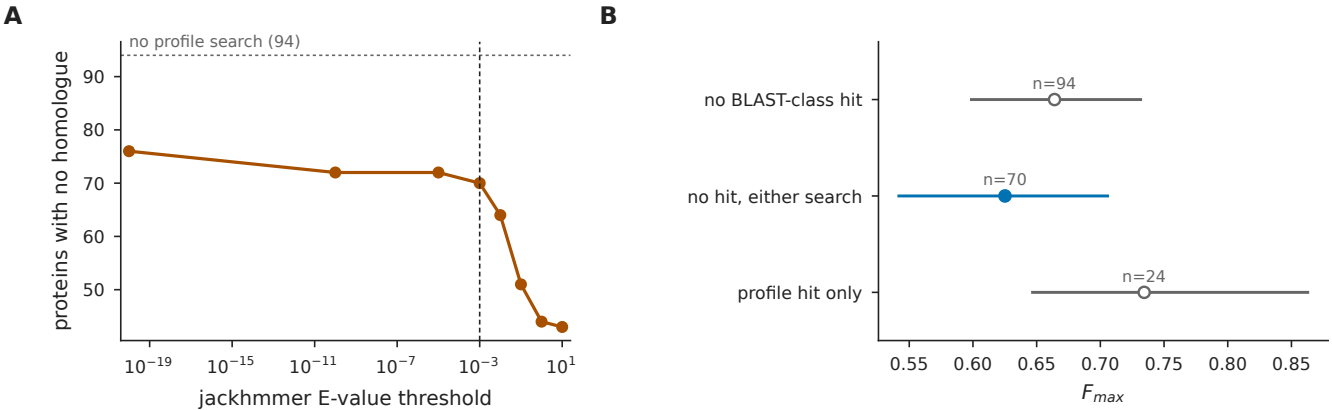The orphan subset withstands a more sensitive search